Jagged AI in Scientific Peer Review: Evidence from POMP Data Analysis

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Abstract

Despite their growing use in academic writing and statistical analysis, the performance of artificial intelligence (AI) tools in scientific peer review remains a largely unexplored area. A key challenge is *jagged AI*, a phenomenon where AI exhibits strong ability spikes in some domains while remaining deficient in others. To study this jaggedness in a practical data science context, we considered the task of reviewing partially observed Markov process (POMP) data analyses. POMP models, also known as state-space models or hidden Markov models, are used to fit mechanistic dynamic models to time series data in diverse applications including disease transmission, ecological dynamics, and financial risk assessment. High-quality peer review in this area entails assessment of scientific context, identification of errors in implementing complex algorithms, and decisions concerning methodological best practices. We studied 72 POMP projects from four semesters of a University of Michigan graduate time series course for which the project reports, the source code, and student peer reviews are anonymized and open-access. We compared the human reviews with four AI reviewing agents, using Claude Code with differing instructions implemented as skill files. We found that AI reviewers exhibited a jagged capability profile, proficiently catching human-overlooked technical errors and invalid inference methodology, while failing to match human standards in checking interpretive errors, narrative coherence, and domain-informed model critique. The jaggedness was found to be similar for all agents, consistent with it being primarily a property of the underlying AI model rather than the specific instructions. Skill file configuration shifted which weaknesses agents emphasized, without removing the jaggedness.

Keywords: Peer review; Partially observed Markov process; jagged artificial intelligence.

1 Introduction

The use of large language models (LLMs) in academic contexts has grown substantially, raising important questions about where AI assistance enhances scientific outcomes and where it degrades them (Vaccaro, Almaatouq, and Malone 2024). A foundational framework for understanding this boundary is the concept of the *jagged technological frontier* (Dell'Acqua et al. 2026). Tasks that appear equally demanding to human knowledge workers can fall on opposite sides of this frontier: where AI capabilities align with the task, AI complements and amplifies human performance; where they do not, AI output becomes unreliable and can actively degrade the quality of work. The benefit of AI assistance is therefore contingent on whether a given task lies within the current capability boundary—a boundary that is neither smooth nor predictable. Jaggedness is not a uniform capability ceiling but a pattern of uneven performance across domains. Morris et al. (2026) characterize it as a tendency for AI to exhibit sharp ability spikes in certain areas while remaining persistently deficient in others, arguing that this unevenness fundamentally complicates any simple notion of AI generality.

Data science peer review represents a complex, multi-layered task that is susceptible to jagged AI performance. A reviewer must verify statistical correctness, evaluate model assumptions, critique scientific reasoning, assess presentation quality, and apply domain knowledge—capabilities that vary considerably in how well they lend themselves to LLM evaluation. Early empirical studies have begun to map where LLM reviewing succeeds and fails: Liu and Shah (2023) found that GPT-4 detects deliberately inserted errors in fewer than half of tested manuscripts, while Liang et al. (2024) showed that AI-generated feedback on papers from major journals overlaps with human reviewers at rates comparable to inter-reviewer agreement, yet tends toward generic rather than targeted critique. These studies focus on AI reviewing of general scientific manuscripts; the performance of AI on technical, domain-specific statistical analyses, where correctness depends on both code implementation and inference methodology, remains largely unexplored. The task of reviewing Partially Observed Markov Process (POMP) data analysis (King, Nguyen, and Ionides 2016) provides a demanding scenario, requiring specialized knowledge of likelihood-based inference, particle filter diagnostics, computationally intensive algorithms, and the interplay between mechanistic model structure and statistical identifiability.

In this paper, we assess the jaggedness of AI in peer reviewing POMP data analysis projects, asking two related questions: (1) Which categories of weaknesses can AI effectively identify in POMP analyses, and which does it consistently miss? (2) Can this jaggedness be tuned through the use of specialized instructions, templates, and scripts for specific workflows? We evaluated four Claude Code agents (i.e., instances of Claude Code provided with specific skill files) across 72 open-access POMP projects, comparing their reviews against graduate student peer reviews validated by the course instructor. The resulting evidence provides a detailed characterization of AI jaggedness in a rigorous statistical peer review context, demonstrating that AI can effectively complement, but not replace, human judgement in scientific evaluation.

2 Materials and Methods

We used a corpus of 72 POMP analysis projects from the STATS 531 course "Modeling and Analysis of Time Series Data" at the University of Michigan, spanning four semesters: Winter 2021 (W21, 16 projects), Winter 2022 (W22, 23 projects), Winter 2024 (W24, 16 projects), and Winter 2025 (W25, 17 projects). The course was not offered in Winter 2023. For each final project, a group of 2-3 graduate students chose a novel scientific scenario relevant to POMP data analysis, found and analyzed data, and wrote a reproducible report. All project reports are available online, anonymously, together with source code. Each project was subjected to blind peer review by approximately 5 students. The students had access to the report and source code. The instructor evaluated each student review, added their own feedback, and posted a final aggregated review online. Student reviews were graded on their ability to identify project weaknesses, but the grades and individual reviews are not publicly available.

We developed four AI agents using Anthropic's Claude framework, each constructed as a specialized AI assistant with a custom system prompt and specific tool access (Anthropic 2025). All reviews were generated using Claude Sonnet 4.6 during April–May 2026. All agents had no internet access or persistent memory, ensuring that each review was conducted independently without cross-contamination between projects. Agents were given Bash, file search, and read tool access, allowing them to search, read, and execute code within the project folder. Each agent was tasked with constructing a structured peer review for every project, producing a list of "major" and "minor" weaknesses as output. Agents were asked to make no more than 15 points, to be comparable to the human lists that typically had 10-12 points. This cap ensures that the agent cannot obtain a high Human Overlap score by generating a large output volume. The four agents differed in their skill file configuration, summarized below. The skill files, additional details of the experimental procedures, and the results obtained, are available at https://github.com/ionides/jagged-ai-peer-review.

Baseline: A baseline agent with no access to skill files, representing the inherent jaggedness of Claude's base model in POMP peer review.

CourseGuided: This agent was equipped with the `guided-pomp-review` skill file, which provided a 13-item checklist grounded in the advice of Wheeler et al. (2024) for generating rigorous, evidence-based peer reviews of POMP manuscripts, covering likelihood inference, benchmark comparisons, convergence diagnostics, and related evaluation criteria. It additionally had access to a `531-references` skill, compiled from past STATS 531 course materials, providing documented course conventions and a catalogue of common errors students were explicitly taught to avoid.

MetaSkill: This agent used the `guided-pomp-review` skill alongside a meta-skill that enabled it to create a new skill file upon discovering a reusable workflow or method of analysis during the review process. This configuration was designed to assess whether AI could iteratively evolve its own context with minimal human intervention. Unlike the other three agents which reviewed projects in parallel, MetaSkill was run sequentially across all 72 projects in order, so that skill files generated during earlier reviews were available to inform later ones.

Orchestrator: This agent loaded the same `guided-pomp-review` skill file as CourseGuided, but applied it within a self-contained multi-step pipeline: an initial review, a dual audit checking evidence grounding and methodological coverage, and a challenge-judge step that stress-tested each flagged point. Only claims that survived the challenge process appeared in the final review. Procedures where an agent orchestrates a multi-stage evaluation process have been found to improve outcomes for LLM annotations in learning analytics (Ahtisham et al. 2026).

Each AI finding was matched to a human issue if they referred to substantially the same underlying concern, even if phrased differently. The major/minor distinction (A vs. C, B vs. D) was determined solely by the AI reviewer's own label. Each human issue was counted exactly once. Each finding was classified into one of six categories:

Table 1: Classification codes for issues raised in review

Code	Description
A	AI major finding — human did not raise
B	AI major finding — human also raised
C	AI minor finding — human did not raise
D	AI minor finding — human also raised
E	Human raised — AI did not address
F	Direct contradiction (excluded from analysis)

The classification was carried out by a team of two agents, called the Comparator and the ComparatorReviewer. The Comparator extracted all distinct concerns from the aggregated human review as a numbered list, then called a new instance of ComparatorReviewer once per AI review agent (Baseline, CourseGuided, MetaSkill and Orchestrator) to classify that reviewer's findings against the human list. This design prevents cross-reviewer contamination: when all four reviewers were analyzed together in a single agent call, the classification agent would misattribute findings between reviewers. We manually checked the AI reviews and classification for all W21 projects and found no entirely unreasonable assessments; major weaknesses in particular were consistently well-grounded. We therefore treated all AI findings as plausible for the remaining semesters. A peer review, carried out by human or AI, that finds some substantive issues can be helpful whether or not other points are correct, and so our conclusions should be insensitive to a small fraction of undetected errors. The primary quantitative metric was *Human Overlap* defined as

$$\text{Human Overlap} = \frac{B + D}{B + D + E}.$$

This measured the fraction of human-confirmed weaknesses independently identified by the AI agent. To assess whether Human Overlap differed across the four agents, we used the Friedman test (Friedman 1937), a non-parametric repeated measures test, treating the 72 projects as repeated units. The null hypothesis H_0 is that the distribution of Human Overlap is the same across all agents, against the alternative H_a that at least one agent differs. Since the aggregated human reviews represent instructor-verified review points, Human Overlap measures the extent to which AI matches validated human assessment. High Human Overlap is not necessarily desirable if the human reviews are missing important points. We addressed this by asking the Comparator to produce cross-reviewer aggregations identifying which human issues no AI reviewer addressed and which AI findings were raised universally across all reviewers. Category E findings were further classified into thematic categories by a separate Claude instance, and then manually verified across all semesters.

3 Results

Table 2 presents the Human Overlap by agent and semester.

Table 2: Human Overlap (%) by agent and semester.

Agent	W21	W22	W24	W25
Baseline	33.3	31.5	22.5	21.1
CourseGuided	33.7	37.1	30.5	23.6
MetaSkill	30.1	35.4	31.1	23.6
Orchestrator	28.9	38.8	28.0	19.0

Figure 1 shows the Human Overlap trend across semesters for each agent. All agents exhibited a declining trend in Human Overlap from W22 to W25, partly because human reviews became more thorough over time: the average number of human-confirmed issues per project grew from 5.2 in W21 to 9.4 in W24 before settling at 8.7 in W25. As the denominator of Human Overlap increased while AI coverage remained roughly stable, the overlap rate naturally declined.

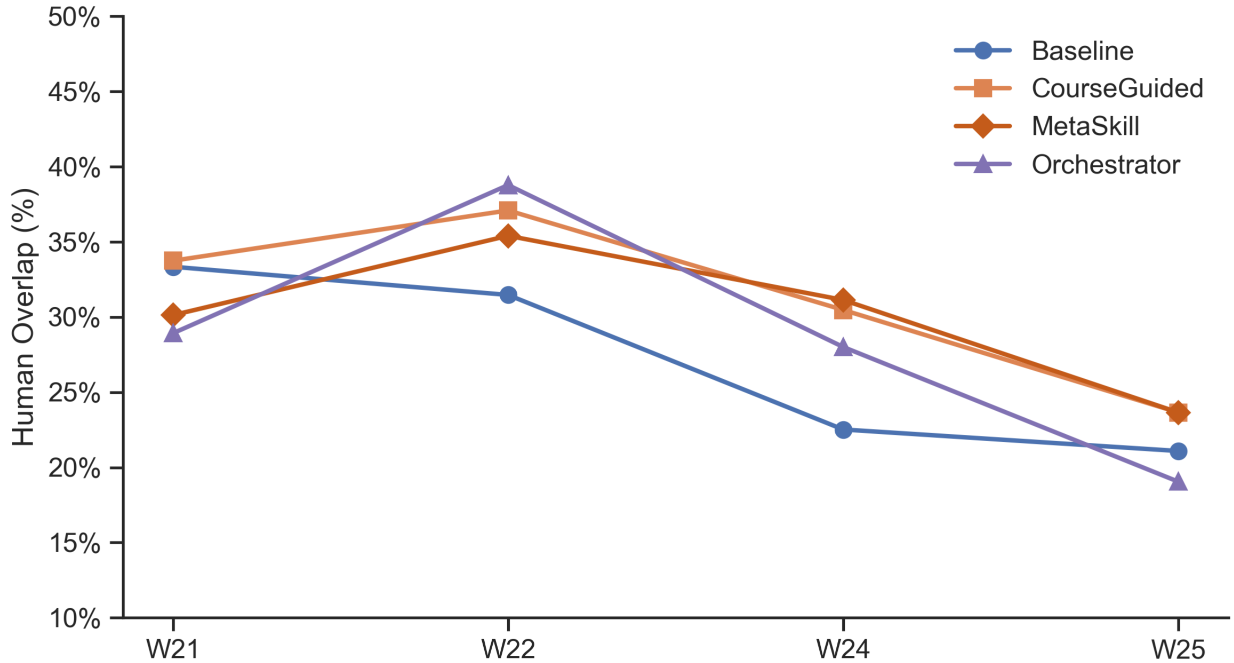

Figure 1: Human Overlap by agent across four semesters. All agents showed declining coverage in later semesters, partly due to increasing human reviewer thoroughness.

The Friedman test on Human Overlap across the four agents yielded p = 0.06, suggesting inclusive evidence against the null hypothesis. The overall mean overlap rates of 29.0% (Baseline), 33.4% (CourseGuided), 31.6% (MetaSkill), and 31.6% (Orchestrator) were similar across agents, with substantial project-level variance visible in Figure 2.

A consistent finding across all agents and semesters was that the majority of each agent's output consisted of AI-unique findings that human reviewers never raised. The Baseline agent generated approximately 6–7 major findings per project that the human reviewer did not raise, while the human reviewer raised 5–9 issues per project that no AI reviewer addressed.

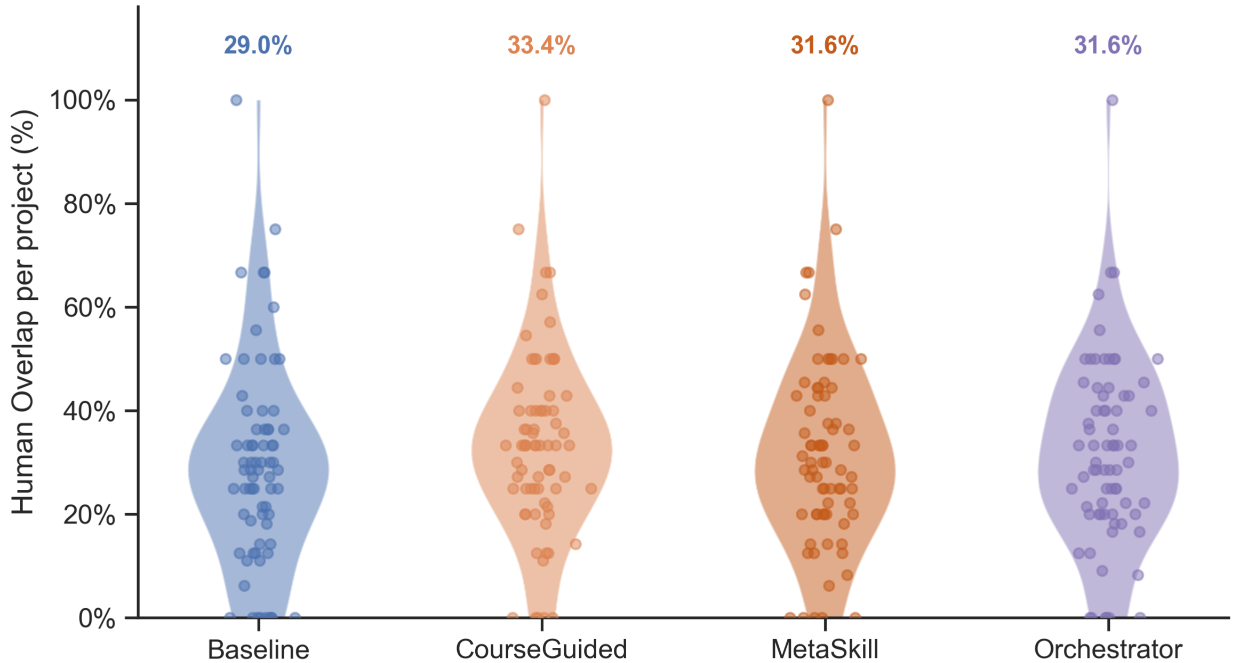

Figure 2: Per-project Human Overlap across all 72 projects for each agent.

3.1 What AI Caught That Humans Missed

Despite low Human Overlap, AI agents generated a large and qualitatively consistent set of findings not raised by human reviewers. The findings divided along the axis of skill file configuration. The Baseline agent proficiently detected errors that required reading and executing code, rather than evaluating the statistical argument. This required the AI to understand particle filters, maximum likelihood via the iterated filtering (IF2) algorithm (Ionides et al. 2015), and implementation via the pomp R package (King, Nguyen, and Ionides 2016). These technical issues fell into four recurring categories, described below.

Code-level bugs that silently corrupt computation. The most consequential class of Baseline-unique findings involved code that ran without error while computing the wrong quantity. Examples included: accumulator variables tracking the wrong quantity (e.g., a compartment in a disease transmission model accumulating recoveries while the measurement model used it as a proxy for new infections), incorrect time-step specifications (e.g., `euler()` instead of `discrete_time()`, introducing spurious sub-steps), and particle filters inadvertently run on simulated rather than real data—rendering the reported benchmark log-likelihood meaningless.

Data handling and series preparation errors. The Baseline agent consistently caught data loaded in reverse chronological order, frequency parameters miscalibrated to the wrong periodicity (e.g., `frequency=365` for trading-day data with approximately 252 trading days per year), silently dropped observations, and discrepancies between parameter values stated in text versus implemented in code.

Search configuration failures. The Baseline agent identified search boxes too narrow to contain the true MLE, search box designs derived from local search results, and ad hoc parameter choices lacking statistical justification.

Reproducibility failures. Undefined variables that would cause runtime errors on re-execution, and hardcoded results in the report that do not match the output of the submitted code.

In contrast to the baseline agent, the skill-equipped agents (CourseGuided and MetaSkill) shifted focus toward inference-methodology completeness, with recurring examples including:

Missing or invalid benchmark comparisons. The CourseGuided agent raised the absence of non-mechanistic baseline time series model comparisons (ARIMA, ARMA, GARCH) as a standalone major finding in approximately 13 of 16 W21 projects. The Baseline agent raised this in none.

Profile likelihood validity. The skill-equipped agents identified specific mechanisms that reduced the validity of profile likelihood analyses. Examples include exclusion of parameters from the iterated filtering random-walk perturbation specification, maximum likelihood estimates lying at search box boundaries, and confidence interval construction based on local rather than global search.

Global search initialization anti-pattern. Both skill-equipped agents systematically flagged the practice of initializing global likelihood searches from prior search results. This approach may be an acceptable practice in some situations, but risks overstatement when described as a global search. The Baseline agent missed this anti-pattern in the majority of cases.

IF2 configuration completeness. The skill-equipped agents audited which parameters were actively perturbed during IF2 optimization, catching cases where key epidemiological parameters were left unperturbed and therefore could not converge to their maximum likelihood estimates.

3.2 What Humans Found That AI Consistently Missed

To characterize the structure of human-unique findings, we analyzed np.int64(152) human-raised issues that were missed (category E) for all four agents. Themes were identified by having Claude broadly classify each finding by content into general categories. The classification was manually verified across all semesters by reviewing the individual assignments and confirming the categorizations were reasonable. These clustered into five recurring themes, shown in Figure 3.

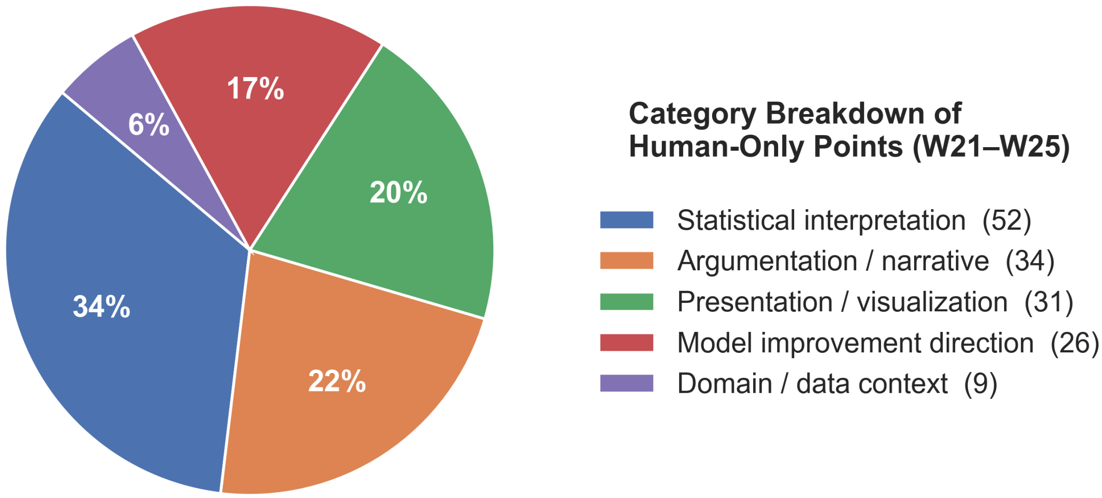

Figure 3: Category breakdown of human-only findings across all semesters (W21–W25). Statistical interpretation accounts for the largest share, followed by argumentation/narrative and presentation/visualization.

Statistical and scientific interpretation (n = 52, 34%). Human reviewers correctly diagnosed the meaning of statistical results in ways AI consistently failed to replicate. Examples included: identifying flat profile likelihoods as evidence of non-identifiability rather than optimization failure, recognizing that bimodal likelihood surfaces warrant investigation and scientific discussion, and catching logically inconsistent conclusions such as rejecting a null hypothesis and then drawing a positive inference from the same p-value.

Argumentation and narrative coherence (n = 34, 22%). Human reviewers assessed whether the scientific argument was internally consistent, the research motivation was clearly stated, the model class was appropriate for the question posed, conclusions followed from the evidence presented, and the analysis contributed insight beyond prior work. The AI agents did not engage with these dimensions of review quality.

Presentation and visualization quality (n = 31, 20%). Human reviewers caught specific figure and table problems that required holistic visual assessment: uninformative or unpolished EDA plots, reversed time axes in figures, excessive significant figures in tables, missing captions and figure numbers, and inconsistent or undefined notation.

Model improvement direction (n = 26, 17%). Human reviewers identified the specific structural change that would resolve an observed problem. Examples included recognizing that weekly periodicity in COVID data necessitates a time-varying measurement model or 7-day differencing, that multiple epidemic waves require additional model compartments rather than parameter tuning, or that over-dispersed process noise would improve fit. AI agents identified that models failed without proposing actionable structural remedies.

Domain and data context (n = 9, 6%). Human reviewers applied domain knowledge that went beyond the code. Examples included questioning how a dataset operationalized contested political concepts (e.g., whether an election was "free and fair"), identifying when stock data used unadjusted close prices affected by splits, and recognizing when modeling assumptions conflicted with domain-specific data conventions.

3.3 Skill File Focus Shift

The quantitative equivalence in Human Overlap across agents masked a qualitatively significant divergence in the *types* of findings each agent generated. Figure 4 illustrates this divergence: the Baseline agent captured implementation errors at consistently high rates, while skill-equipped agents systematically flagged inference-methodology violations that the Baseline missed.

Domain Specific Breakdown of Agent-Unique Review Overlap	Baseline	CourseGuided / MetaSkill	Orchestrator
AI-unique findings (A+C) — total across W21–W25	929 (12.9/proj)	958 (13.3/proj)	641 (8.9/proj)
Global search inherits cooled schedule, not truly global (optimization flaw)	✓	✓	✓
No ESS or particle filter diagnostics reported (verification gap)	✓	✓	✓
Cross-family log-likelihood comparison invalid (scale mismatch)	✓	✓	✓
Modifying one variable silently changes another (implementation bug)	✓		
Code does not implement the model as written (implementation bug)	✓		
Computation produces wrong values without error (implementation bug)	✓		
Confidence interval procedure is wrong (methodology flaw)		✓	
No simpler baseline model for comparison (model evaluation)		✓	
Too few starting values explored in fitting (optimization flaw)		✓	
Profile too flat/noisy to extract CIs (identifiability issue)			✓
Incorrect biological formula in model (domain error)			✓
Results lack parameter estimates or captions (omission)			✓

Figure 4: Domain-specific comparison of agent-unique findings across all semesters. Checkmarks indicate findings systematically raised by that agent class but not others. The Baseline agent excelled at implementation checks; skill-equipped agents redirected focus to inference methodology.

This trade-off was directional and category-specific. The Baseline agent focused on data, code, and reporting errors: particle filter object identity, silent computation corruption, text-code mismatches at the parameter level, and attribution failures. The skill-equipped agents' focused on inference-methodology completeness: benchmark comparison requirements, profile likelihood validity, global search initialization correctness, and model diagnostic outputs, following the context framed by the skill files. The trade-off is consistent with the Friedman test result: the skill files redirected the agent to a different category of points without increasing total overlap with the human-identified weaknesses. CourseGuided and MetaSkill are grouped together (and their totals averaged) in Figure 4 because their qualitative finding patterns were nearly identical despite MetaSkill accumulating additional skill files throughout its sequential run. This similarity reflects a point of diminishing returns: beyond a certain point, adding context through skill files did not produce meaningfully different coverage.

4 Discussion

Our results showed that the AI reviewers exhibited a jagged capability profile in POMP peer review. The findings can be organized around three central observations.

1. AI and human reviewers largely detected different problems. Across all four semesters and all agents, fewer than 40% of human-confirmed weaknesses were independently identified by any AI reviewer. The majority of AI findings were entirely absent from human reviews, while most human-identified issues went unaddressed by AI. This complementary structure held regardless of skill file configuration. AI reviewers can serve as a valuable supplement to human review precisely because they cover ground that human reviewers—who typically do not read and execute source code—would not otherwise check.

2. The addition of skill files tuned jaggedness rather than resolving it. When equipped with a domain-specific skill file, the agent's focus pivoted toward inference methodology within the context detailed therein—producing stronger coverage of benchmark comparisons, profile likelihoods, convergence diagnostics, and POMP-specific inference standards. The agent's assigned skills also affected the classification of a given issue as major or minor. However, the skills did not significantly affect the overlap with the human reviewers, and therefore did not fundamentally affect the jaggedness of the review.

3. The human-unique layer was characterized by judgment and context that the AI agents did not supply. The five recurring categories of human-unique findings (statistical interpretation, argumentation, presentation quality, model improvement direction, and domain context) shared a common feature: they required integrating information across the project holistically, applying domain knowledge that went beyond the text, and exercising scientific judgment about what questions and answers are worthwhile. Across the different skill contexts, these AI agents did not raise these issues. For example, the AI reviewers could verify whether a profile likelihood was correctly computed, but not whether the interpretation of a flat profile was scientifically sound. A flat profile likelihood suggests a lack of marginal information about a parameter, and this could be problematic, or could be just a fact of marginal relevance if other interesting parameters show curvature in their respective profile likelihoods, or could suggest reparameterizing to look for a well-identified parameter combination. Some diligent human referees were able to address these interpretive issues. As another example, the AI agents could detect a missing benchmark comparison, but not whether the model class was fundamentally inappropriate for the scientific question.

Our results cannot necessarily be extrapolated to other agent configurations or other foundation models. The positive finding, that our AI review agents identified useful insights consistently missed by humans, is robust to this limitation. The negative finding, that our AI review agents had significant limitations, could be reassessed by future research. Such limitations will exist as long as AI capabilities remain jagged, but the boundary of the limitations will change. A game-theoretic analysis of jaggedness suggests that it places high value on understanding the specific capabilities and limitations of AI (Gans 2026), advocating for the value of benchmarking studies such as this.

5 Conclusions

We evaluated four Claude agents with varying skill file configurations on 72 POMP analysis projects, comparing their output to open-access human peer reviews. The central finding was that AI reviewers exhibited a jagged capability profile: they proficiently caught technical implementation errors and invalid inference methodology that human reviewers often overlooked, while consistently failing to match human standards in statistical interpretation, narrative coherence, model improvement direction, and domain-informed critique. The addition of domain-specific guidance, via skill file configurations, shifted the agent's focus without affecting jaggedness, leaving the overall proportion of human-confirmed weaknesses detected statistically unchanged across all four agent configurations.

The complementarity between AI and human review, with each detecting a largely non-overlapping set of problems, suggests that AI review can currently serve as a valuable supplement to human review but not as a replacement for human judgement. AI is particularly well-suited to catching implementation-layer errors that human reviewers may not have the time to identify. The use of AI to develop scientific data analysis will hasten the use of complex methodology that cannot readily be fully evaluated without AI assistance. Scientific journals will soon need to accept a role for AI in peer review.

We studied graduate student peer review of course final projects, which is a more controlled environment than journal peer review. In particular, students were required to submit source code that reproduced the project report, using R markdown. The AI agents may have benefitted more than the humans from this high expectation for reproducibility. Another difference is that final projects may have more mistakes, since they represent around two weeks of work. However, we postulate that mistakes made by statistics graduate students after taking an advanced course may be representative of mistakes made by practitioners using similar statistical methods for scientific investigations.

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Ethics. This study analyzes publicly available and de-identified data. A request is pending for exemption from full review by the University of Michigan Institutional Review Board (HUM00293579).

Use of AI. AI tools were used as described in Materials and Methods. Claude Code Opus 4.7 and Chat GPT 5.5 were used to provide review of this manuscript, and we are grateful for their helpful suggestions.

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6 Supplementary Material

Tables Table 3 and Table 4 summarize the full review output of the agents and humans. The complete experimental results are available online (see Materials and Methods).

Table 3: Raw finding counts (A, B, C, D, E, F) by agent and semester.

Table 3

Semester	Agent	A	B	C	D	E	F
W21	Baseline	97	16	113	12	56	0
W21	CourseGuided	101	20	114	8	55	0
W21	MetaSkill	106	19	121	6	58	0
W21	Orchestrator	61	13	93	11	59	1
W22	Baseline	149	33	139	23	122	1
W22	CourseGuided	139	43	140	23	112	1
W22	MetaSkill	155	39	148	24	115	1
W22	Orchestrator	88	35	101	34	109	1
W24	Baseline	100	26	109	8	117	0
W24	CourseGuided	90	37	121	9	105	0
W24	MetaSkill	93	33	116	14	104	0
W24	Orchestrator	54	31	81	11	108	0
W25	Baseline	118	21	104	10	116	1
W25	CourseGuided	108	24	129	11	113	0
W25	MetaSkill	107	25	129	10	113	0
W25	Orchestrator	78	15	85	13	119	1

Table 4: Counts of human-only findings (E category for all agents) by theme across all semesters.

Table 4

Theme	Count	Percentage (%)
Statistical interpretation	52	34.2
Argumentation / narrative	34	22.4

Theme	Count	Percentage (%)
Presentation / visualization	31	20.4
Model improvement direction	26	17.1
Domain / data context	9	5.9